

01_EDA Online Retail exploratory data analysis

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1. Project outline

UCI Online Retail (id 352) one year (Dec 2010 – Dec 2011) of transactions from a UK-based online gift retailer.

dataPreprocess.py writes four files into ../data/:

- online_retail_clean.csv positive-quantity, positive-price line items
- online_retail_cancelled.csv cancellation invoices kept SEPARATELY (so net-of-returns features can be computed without throwing them away)
- customer_features.csv customer-level RFM and behavioural features
- data_quality_summary.txt sanity-check counts

2. Load and validate

```
tx          <- read_csv("../data/interim/online_retail_clean.csv",
                        show_col_types = FALSE)
cancelled   <- read_csv("../data/interim/online_retail_cancelled.csv",
                        show_col_types = FALSE)
customer    <- read_csv("../data/processed/customer_features.csv",
                        show_col_types = FALSE)

tx$InvoiceDate <- as_datetime(tx$InvoiceDate)
tx$Date        <- as_date(tx$Date)
cancelled$InvoiceDate <- as_datetime(cancelled$InvoiceDate)

# Standardise CustomerID (capital ID) and clean Description encoding
# BEFORE any plot or table needs them.
tx <- tx |>
  rename_with(~(n) ifelse(n == "CustomerId", "CustomerID", n)) |>
  mutate(Description = iconv(Description, from = "", to = "UTF-8", sub = ""))

# Validation fail fast if data shape isn't what later chunks assume.
stopifnot(
  "CustomerID" %in% names(tx),
  "CustomerID" %in% names(customer),
  !anyDuplicated(customer$CustomerID),
  nrow(customer) > 0,
```

```

nrow(tx) > 0
)

list(
  transactions = dim(tx),
  cancellations = dim(cancelled),
  customers = dim(customer),
  date_range = range(tx$InvoiceDate),
  countries = n_distinct(tx$Country)
)

## $transactions
## [1] 391150      14
##
## $cancellations
## [1] 8506      9
##
## $customers
## [1] 4334     29
##
## $date_range
## [1] "2010-12-01 08:26:00 UTC" "2011-12-09 12:50:00 UTC"
##
## $countries
## [1] 37

```

```
glimpse(customer)
```

```

## Rows: 4,334
## Columns: 29
## $ CustomerID      <dbl> 12346, 12347, 12348, 12349, 12350, 12352, 12353, 1~
## $ Recency          <dbl> 326, 2, 75, 19, 310, 36, 204, 232, 214, 23, 33, 2,~
## $ Frequency        <dbl> 1, 7, 4, 1, 1, 7, 1, 1, 1, 3, 1, 2, 4, 3, 1, 10, 2~
## $ GrossMonetary     <dbl> 77183.60, 4310.00, 1437.24, 1457.55, 294.40, 1385.~
## $ TotalQuantity     <dbl> 74215, 2458, 2332, 630, 196, 526, 20, 530, 240, 15~
## $ UniqueProducts    <dbl> 1, 103, 21, 72, 16, 57, 4, 58, 13, 52, 131, 12, 21~
## $ FirstPurchase     <dtm> 2011-01-18 10:01:00, 2010-12-07 14:57:00, 2010-12~
## $ LastPurchase      <dtm> 2011-01-18 10:01:00, 2011-12-07 15:52:00, 2011-09~
## $ TenureDays         <dbl> 0, 365, 282, 0, 0, 260, 0, 0, 0, 302, 0, 149, 274,~
## $ SinglePurchase     <dbl> 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0,~
## $ AvgUnitPrice       <dbl> 1.0400000, 1.7534581, 0.6163122, 2.3135714, 1.5020~
## $ AvgOrderValue      <dbl> 77183.6000, 615.7143, 359.3100, 1457.5500, 294.400~
## $ AvgOrderItems      <dbl> 74215.00000, 351.14286, 583.00000, 630.00000, 196.~
## $ ReturnInvoices     <dbl> 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 2, 0, 0, 3, 0,~
## $ ReturnValueAbs     <dbl> 77183.60, 0.00, 0.00, 0.00, 0.00, 120.33, 0.00, 0.~
## $ Country           <chr> "United Kingdom", "Iceland", "Finland", "Italy", "~
## $ NetMonetary        <dbl> 0.00, 4310.00, 1437.24, 1457.55, 294.40, 1265.41, ~
## $ ReturnRate         <dbl> 0.5000000, 0.0000000, 0.0000000, 0.0000000, 0.0000~
## $ ReturnValueRate    <dbl> 1.000000000, 0.000000000, 0.000000000, 0.000000000~
## $ AvgUnitPrice_winsor <dbl> 1.0400000, 1.7534581, 0.6163122, 2.3135714, 1.5020~
## $ log_Recency        <dbl> 5.789960, 1.098612, 4.330733, 2.995732, 5.739793, ~
## $ log_Frequency      <dbl> 0.6931472, 2.0794415, 1.6094379, 0.6931472, 0.6931~
## $ log_GrossMonetary  <dbl> 11.253955, 8.368925, 7.271175, 7.285198, 5.688330,~

```

```
## $ log_TotalQuantity    <dbl> 11.214735, 7.807510, 7.754910, 6.447306, 5.283204,~
## $ log_UniqueProducts   <dbl> 0.6931472, 4.6443909, 3.0910425, 4.2904594, 2.8332~
## $ log_AvgOrderValue    <dbl> 11.253955, 6.424406, 5.886965, 7.285198, 5.688330,~
## $ log_NetMonetary      <dbl> 0.000000, 8.368925, 7.271175, 7.285198, 5.688330, ~
## $ Monetary            <dbl> 77183.60, 4310.00, 1437.24, 1457.55, 294.40, 1385.~
## $ log_Monetary         <dbl> 11.253955, 8.368925, 7.271175, 7.285198, 5.688330,~
```

3. Headline numbers gross AND net

The pipeline now reports both gross sales and net-of-cancellations revenue. The difference is the share of sales customers later cancelled.

```
gross_rev    <- sum(tx$Amount)
return_value <- sum(abs(cancelled$Amount))
net_rev      <- gross_rev - return_value

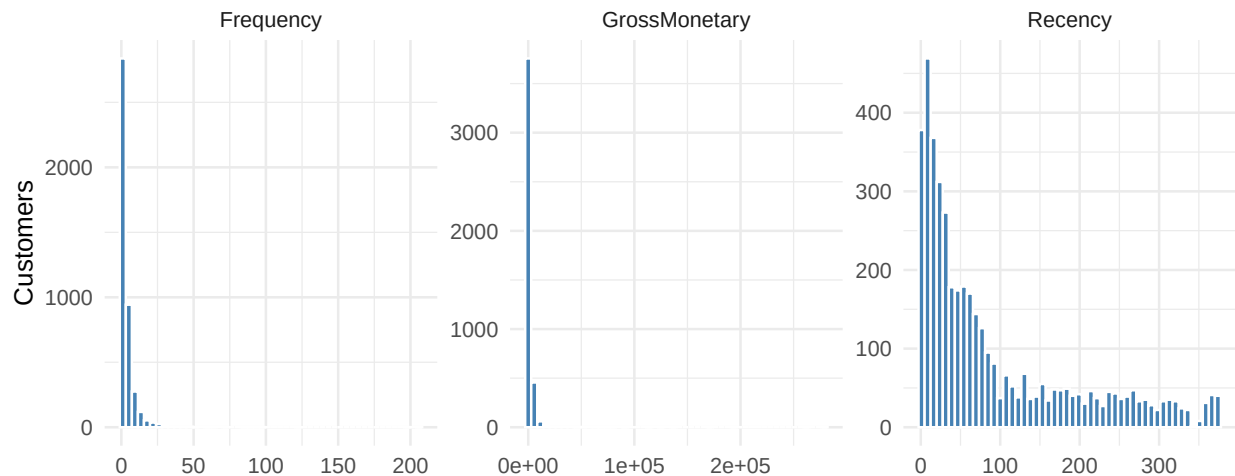
tibble(
  customers      = n_distinct(tx$CustomerID),
  invoices       = n_distinct(tx$InvoiceNo),
  cancelled_invoices = n_distinct(cancelled$InvoiceNo),
  unique_products = n_distinct(tx$StockCode),
  line_items     = nrow(tx),
  gross_revenue  = gross_rev,
  return_value_abs = return_value,
  net_revenue    = net_rev,
  return_value_rate = return_value / gross_rev
) |>
pivot_longer(everything(), names_to = "metric", values_to = "value") |>
mutate(value = case_when(
  str_detect(metric, "rate") ~ percent(value, accuracy = 0.1),
  TRUE                        ~ comma(value, accuracy = 0.01)
))
```

```
## # A tibble: 9 x 2
##   metric      value
##   <chr>      <chr>
## 1 customers  4,334.00
## 2 invoices  18,402.00
## 3 cancelled_invoices 3,383.00
## 4 unique_products  3,659.00
## 5 line_items  391,150.00
## 6 gross_revenue  8,737,227.64
## 7 return_value_abs 471,750.71
## 8 net_revenue    8,265,476.93
## 9 return_value_rate 5.4%
```

4. Distribution of RFM features (raw vs. log)

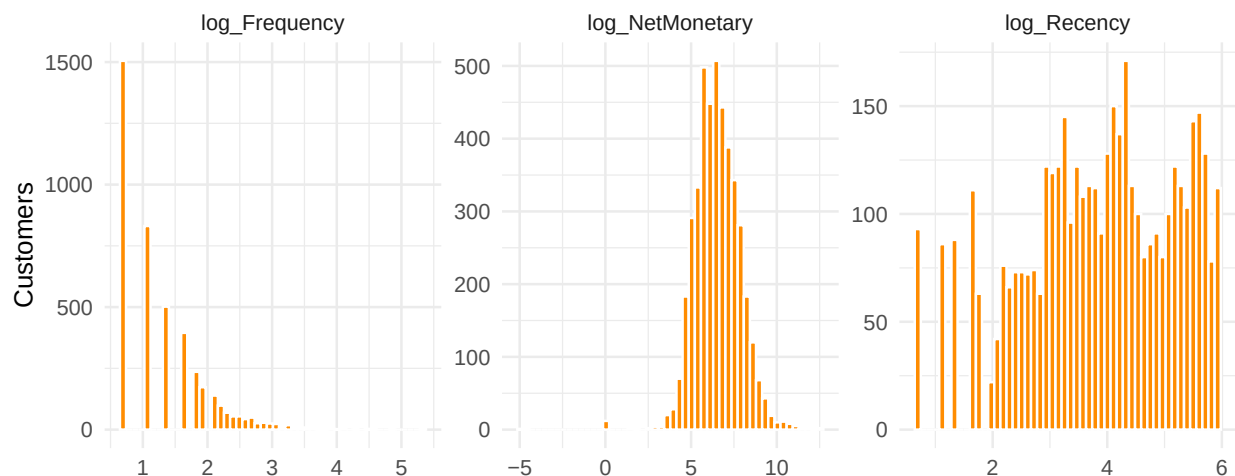
```
customer |>
  select(Recency, Frequency, GrossMonetary) |>
  pivot_longer(everything(), names_to = "feature", values_to = "value") |>
  ggplot(aes(value)) +
  geom_histogram(bins = 50, fill = "steelblue", colour = "white") +
  facet_wrap(~ feature, scales = "free", nrow = 1) +
  labs(title = "Raw RFM distributions (heavy right tail)",
       x = NULL, y = "Customers")
```

Raw RFM distributions (heavy right tail)



```
customer |>
  select(log_Recency, log_Frequency, log_NetMonetary) |>
  pivot_longer(everything(), names_to = "feature", values_to = "value") |>
  ggplot(aes(value)) +
  geom_histogram(bins = 50, fill = "darkorange", colour = "white") +
  facet_wrap(~ feature, scales = "free", nrow = 1) +
  labs(title = "Log-transformed RFM (closer to symmetric, but tail remains)",
       x = NULL, y = "Customers")
```

Log-transformed RFM (closer to symmetric, but tail remains)



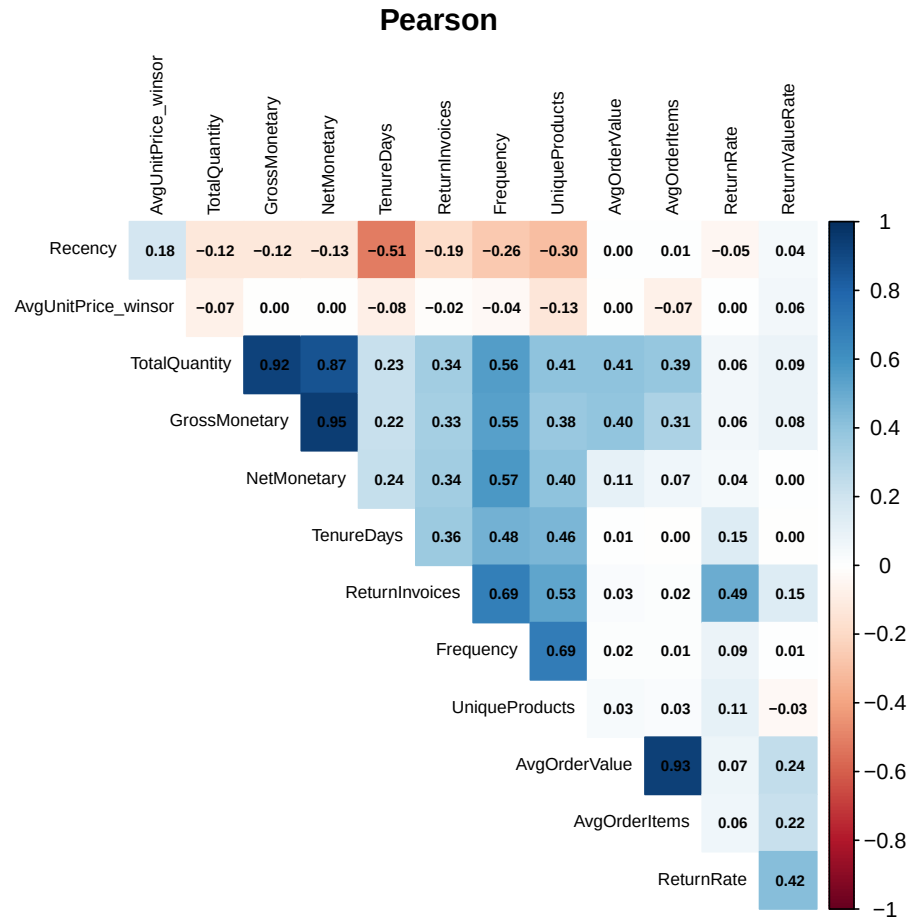
The log transform reduces but does not eliminate skew. `log_NetMonetary` is approximately bell-shaped, but the upper shoulder represents real high-value (likely wholesale-leaning) customers and will continue to influence distance-based clustering.

5. Correlation structure Pearson and Spearman

We report both because Pearson is sensitive to outliers; Spearman (rank-based) is robust to them. Pairs that look correlated under Pearson but not Spearman are correlated because of extreme values useful to know when choosing which features to keep.

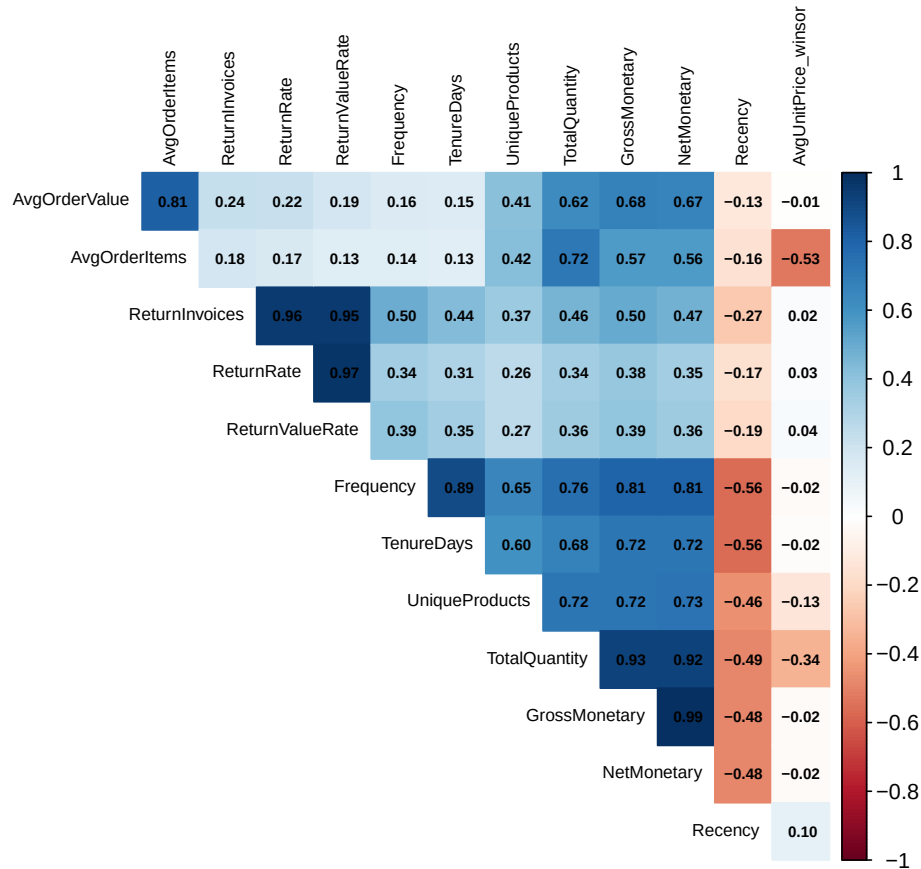
```
num_cols <- customer |>
  select(Recency, Frequency, GrossMonetary, NetMonetary,
         TotalQuantity, UniqueProducts,
         AvgUnitPrice_winsor, AvgOrderValue, AvgOrderItems,
         TenureDays, ReturnInvoices, ReturnRate, ReturnValueRate)

corrplot::corrplot(
  cor(num_cols, use = "pairwise.complete.obs"),
  method = "color", type = "upper", order = "hclust",
  addCoef.col = "black", number.cex = 0.6,
  tl.col = "black", tl.cex = 0.7, diag = FALSE,
  title = "Pearson", mar = c(0, 0, 2, 0)
)
```



```
corrplot::corrplot(
  cor(num_cols, use = "pairwise.complete.obs", method = "spearman"),
  method = "color", type = "upper", order = "hclust",
  addCoef.col = "black", number.cex = 0.6,
  tl.col = "black", tl.cex = 0.7, diag = FALSE,
  title = "Spearman (rank-based)", mar = c(0, 0, 2, 0)
)
```

Spearman (rank-based)



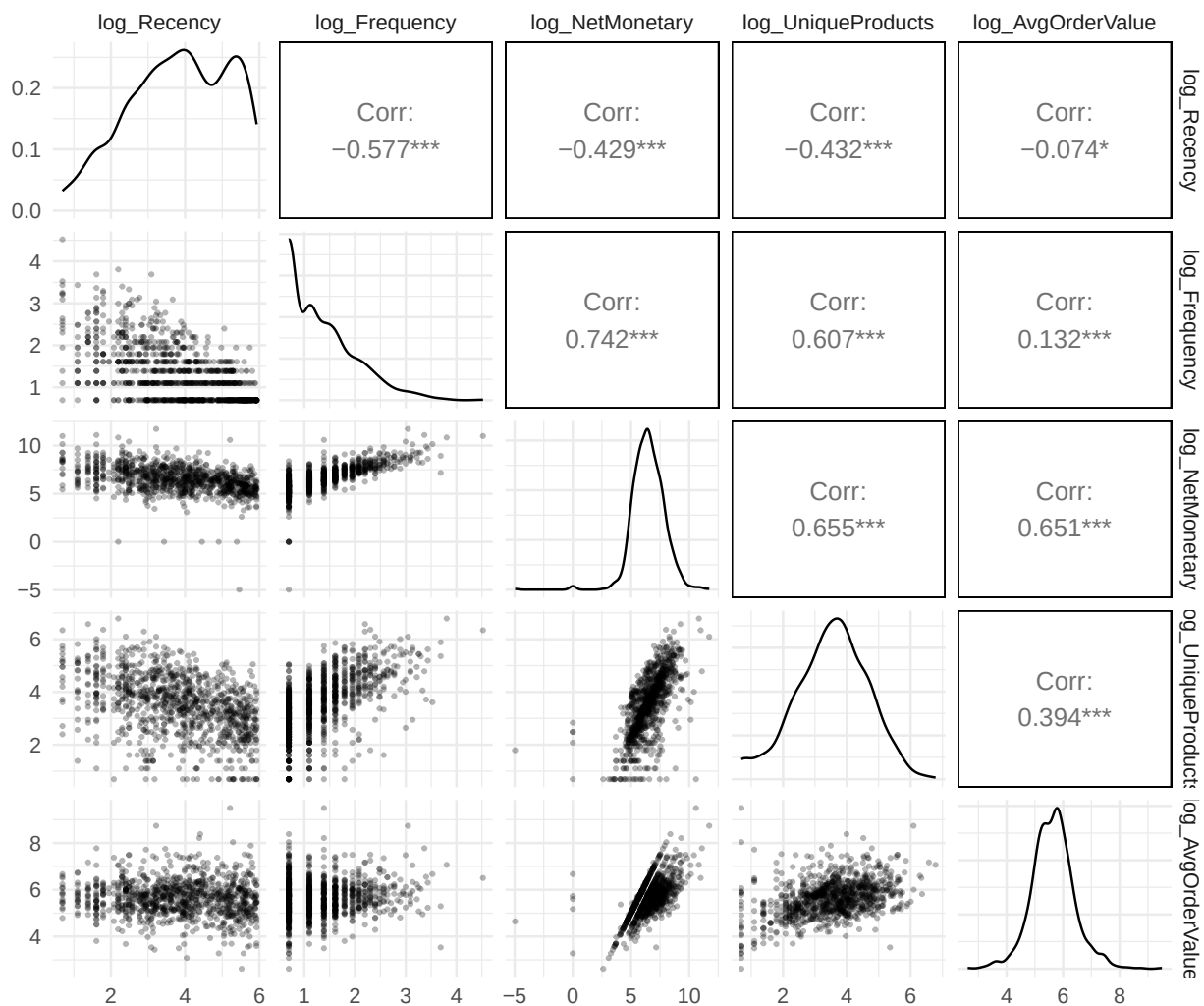
What this implies for clustering input selection:

- **GrossMonetary**, **NetMonetary**, **TotalQuantity**, and **Frequency** form a tight collinear block. We keep **NetMonetary** because it captures the customer's *actual* contribution to revenue net of returns.
- **AvgOrderValue** and **AvgOrderItems** carry similar information; we keep **AvgOrderValue** (revenue-relevant) and drop **AvgOrderItems**.
- **Recency** is largely independent of the spending features under both correlation measures it is its own axis of variation, justifying the canonical R, F, M triple.
- **ReturnRate** and **ReturnValueRate** are weakly correlated with everything else. They are best used as *profiling* variables, not clustering inputs.

6. Pairs plot on the modelling set

```
set.seed(1)
customer |>
  slice_sample(n = 1000) |>
  select(log_Recency, log_Frequency, log_NetMonetary,
         log_UniqueProducts, log_AvgOrderValue) |>
  GGally::ggpairs(progress = FALSE,
                  lower = list(continuous = wrap("points", alpha = 0.3,
                                                  size = 0.6))) +
  labs(title = "Log-feature pairs plot (1k-customer sample)")
```

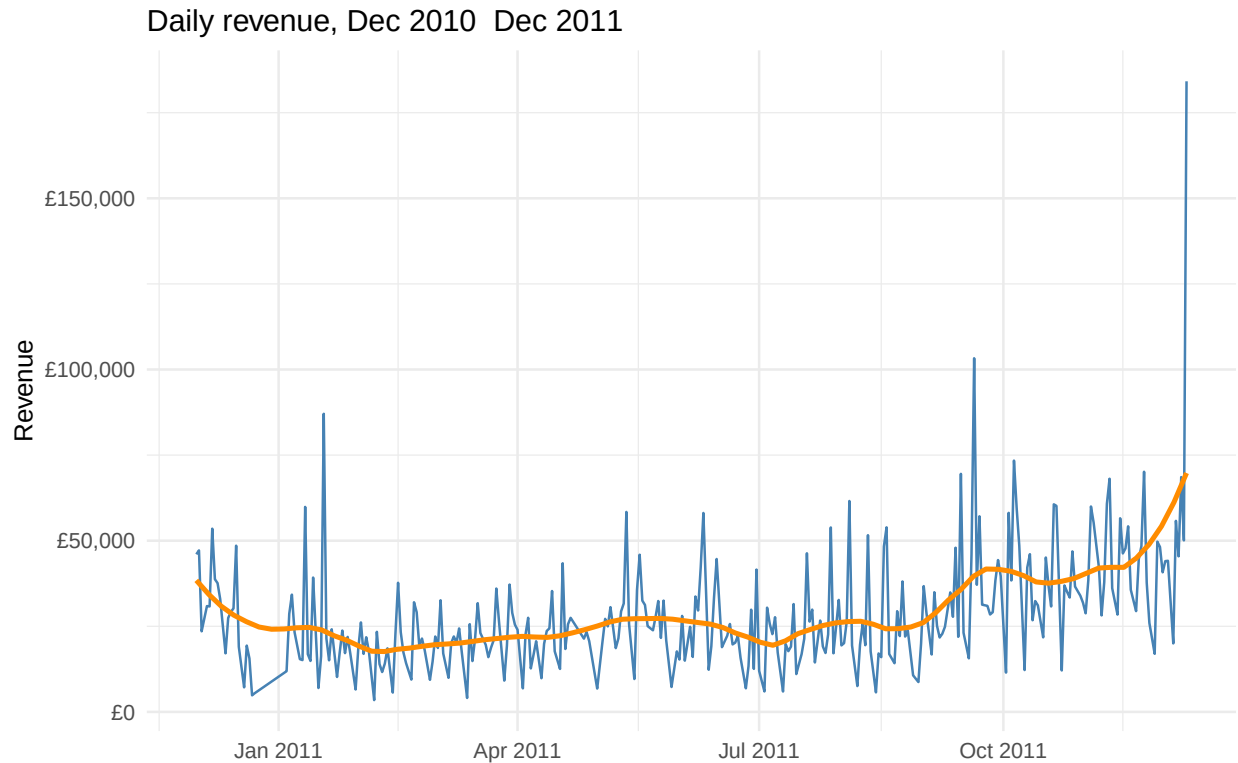
Log-feature pairs plot (1k-customer sample)



7. Sales over time

```
daily <- tx |>
  group_by(Date) |>
  summarise(revenue = sum(Amount),
            invoices = n_distinct(InvoiceNo), .groups = "drop")

ggplot(daily, aes(Date, revenue)) +
  geom_line(colour = "steelblue") +
  geom_smooth(method = "loess", span = 0.2, se = FALSE,
             colour = "darkorange") +
  scale_y_continuous(labels = label_comma(prefix = pound)) +
  labs(title = paste("Daily revenue, Dec 2010", emdash, "Dec 2011"),
       x = NULL, y = "Revenue")
```



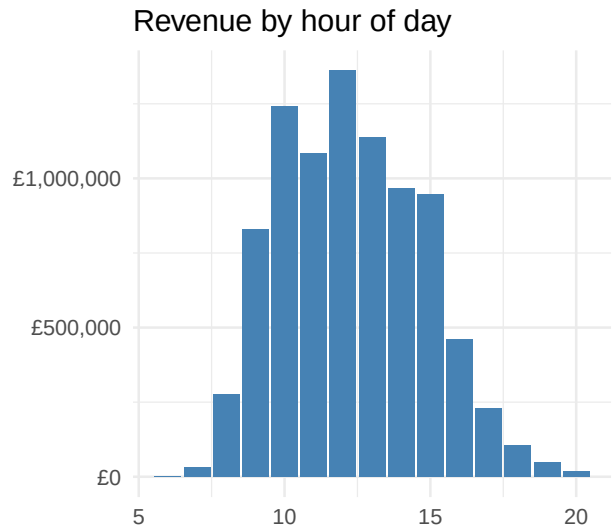
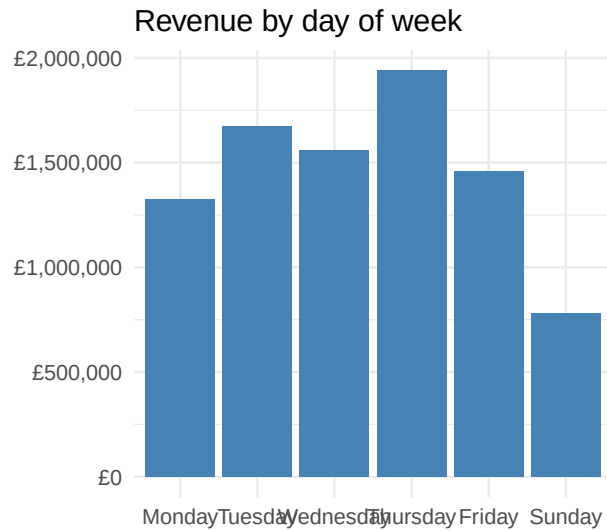
The Q4 2011 ramp is the holiday gift season a sanity check that the timestamps and currency conversions are correct. The single-day spikes in late November / early December correspond to a small number of very large invoices.

```
library(gridExtra)

p_dow <- tx |>
  dplyr::count(DOW = factor(DOW,
    levels = c("Monday", "Tuesday", "Wednesday", "Thursday",
      "Friday", "Saturday", "Sunday")),
    wt = Amount, name = "revenue") |>
  ggplot(aes(DOW, revenue)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(labels = label_comma(prefix = pound)) +
  labs(title = "Revenue by day of week", x = NULL, y = NULL)

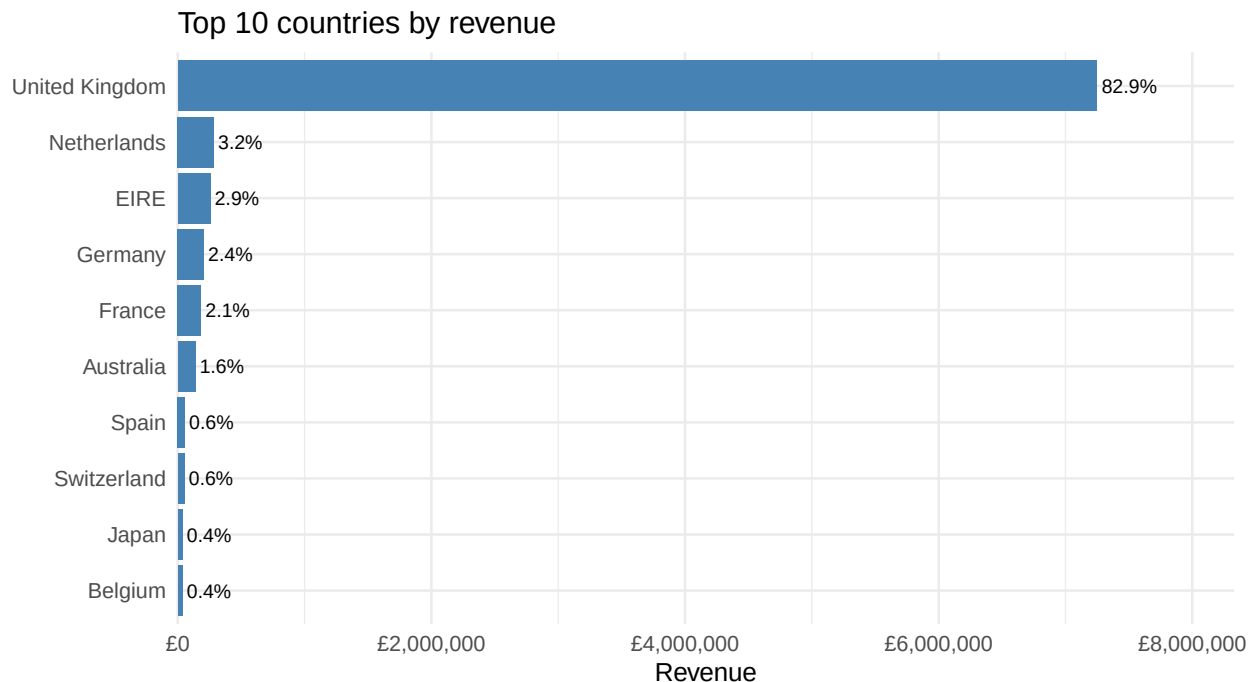
p_hour <- tx |>
  dplyr::count(Hour, wt = Amount, name = "revenue") |>
  ggplot(aes(Hour, revenue)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(labels = label_comma(prefix = pound)) +
  labs(title = "Revenue by hour of day", x = NULL, y = NULL)

gridExtra::grid.arrange(p_dow, p_hour, nrow = 1)
```



8. Geography

```
tx |>
  dplyr::count(Country, wt = Amount, name = "revenue", sort = TRUE) |>
  mutate(
    share = revenue / sum(revenue),
    Country = fct_reorder(Country, revenue)
  ) |>
  slice_max(revenue, n = 10) |>
  ggplot(aes(revenue, Country)) +
  geom_col(fill = "steelblue") +
  geom_text(
    aes(label = percent(share, accuracy = 0.1)),
    hjust = -0.1,
    size = 3
  ) +
  scale_x_continuous(
    labels = label_comma(prefix = pound),
    expand = expansion(mult = c(0, 0.15))
  ) +
  labs(
    title = "Top 10 countries by revenue",
    x = "Revenue",
    y = NULL
  ) +
  coord_cartesian(clip = "off")
```



The UK dominates roughly four-fifths of net revenue, which is why `Country` is held out of the clustering input matrix in `O2_PCA.Rmd`: including it would force a near-degenerate UK / non-UK split that adds no behavioural signal.

9. Anomaly check single-day revenue spikes

The smoothed loess in §7 shows real seasonality, but the residuals reveal a handful of days whose revenue is several standard deviations above the local trend. Distance-based clusterers are sensitive to a small number of extreme observations, so we have to know which invoices and which products drive these spikes before deciding what if anything to do with them.

9a. Days with the largest residual revenue

```
daily_res <- daily |>
  arrange(Date) |>
  mutate(trend = predict(loess(revenue ~ as.numeric(Date),
                              data = pick(everything()),
                              span = 0.2)),
         resid = revenue - trend) |>
  arrange(desc(resid))

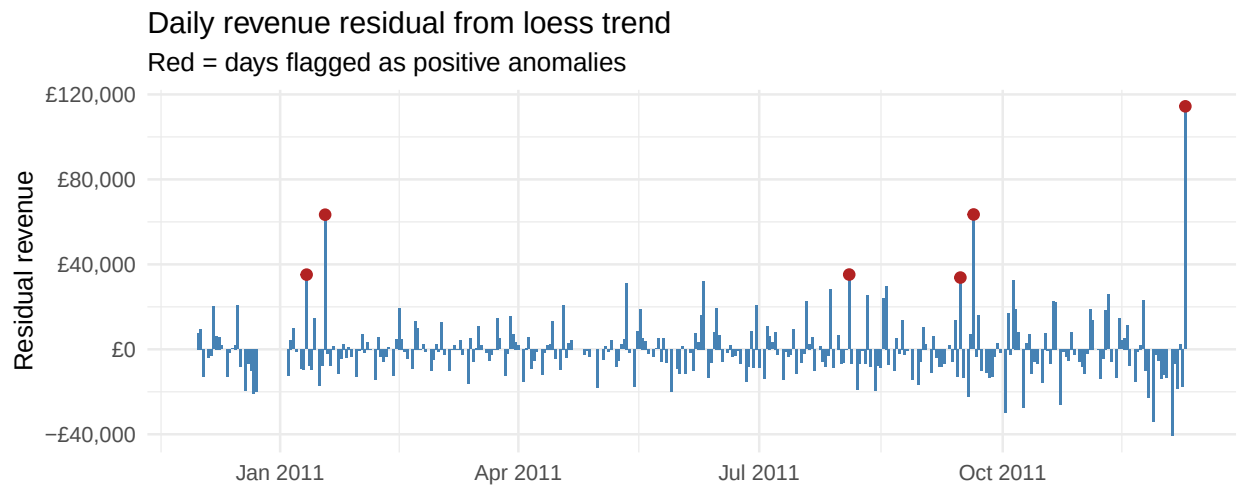
top_days <- daily_res |> slice_head(n = 6)

top_days |>
  mutate(across(c(revenue, trend, resid),
                 \(x) comma(x, accuracy = 1, prefix = pound))) |>
  knitr::kable(caption = "Six days with the largest positive residual revenue")
```

Table 1: Six days with the largest positive residual revenue

Date	revenue	invoices	trend	resid
2011-12-09	£184,171	41	£69,764	£114,407
2011-09-20	£103,280	67	£39,790	£63,490
2011-01-18	£87,066	31	£23,679	£63,387
2011-08-04	£61,600	79	£26,397	£35,203
2011-01-11	£59,865	53	£24,700	£35,165
2011-09-15	£69,521	76	£35,732	£33,788

```
ggplot(daily_res, aes(Date, resid)) +
  geom_col(fill = "steelblue") +
  geom_point(data = top_days, aes(Date, resid),
            colour = "firebrick", size = 2) +
  scale_y_continuous(labels = label_comma(prefix = pound)) +
  labs(title = "Daily revenue residual from loess trend",
       subtitle = "Red = days flagged as positive anomalies",
       x = NULL, y = "Residual revenue")
```



9b. Top contributing invoices on those days

```
top_invoices <- tx |>
  filter(Date %in% top_days$Date) |>
  group_by(Date, InvoiceNo, CustomerID, Country) |>
  summarise(invoice_revenue = sum(Amount),
            line_items      = n(),
            .groups = "drop") |>
  arrange(desc(invoice_revenue)) |>
  slice_head(n = 10)

top_invoices |>
  mutate(invoice_revenue = comma(invoice_revenue, accuracy = 1,
                                prefix = pound)) |>
  knitr::kable(caption = "Top 10 invoices on flagged days")
```

Table 2: Top 10 invoices on flagged days

Date	InvoiceNo	CustomerID	Country	invoice_revenue	line_items
2011-12-09	581483	16446	United Kingdom	£168,470	1
2011-01-18	541431	12346	United Kingdom	£77,184	1
2011-09-20	567423	17450	United Kingdom	£31,698	12
2011-09-20	567381	17450	United Kingdom	£22,105	12
2011-08-04	562439	12931	United Kingdom	£18,841	22
2011-01-11	540815	15749	United Kingdom	£15,161	3
2011-09-15	566934	18102	United Kingdom	£13,250	8
2011-01-11	540689	17450	United Kingdom	£12,798	4
2011-01-11	540818	15749	United Kingdom	£7,838	2
2011-09-15	566935	18102	United Kingdom	£6,870	5

9c. Top contributing products

```

top_products <- tx |>
  filter(Date %in% top_days$Date) |>
  group_by(StockCode, Description) |>
  summarise(revenue = sum(Amount),
            quantity = sum(Quantity),
            invoices = n_distinct(InvoiceNo),
            .groups = "drop") |>
  arrange(desc(revenue)) |>
  slice_head(n = 8)

top_products |>
  mutate(Description = str_trunc(Description, 35),
         revenue = comma(revenue, accuracy = 1, prefix = pound),
         quantity = comma(quantity, accuracy = 1)) |>
  knitr::kable(caption = "Top 8 products on flagged days")

```

Table 3: Top 8 products on flagged days

StockCode	Description	revenue	quantity	invoices
23843	PAPER CRAFT , LITTLE BIRDIE	£168,470	80,995	1
23166	MEDIUM CERAMIC TOP STORAGE JAR	£77,299	74,315	8
85123A	WHITE HANGING HEART T-LIGHT HOLDER	£10,691	3,800	38
23243	SET OF TEA COFFEE SUGAR TINS PANTRY	£7,651	1,526	13
48185	DOORMAT FAIRY CAKE	£6,936	1,194	4
21108	FAIRY CAKE FLANNEL ASSORTED COLOUR	£6,637	3,177	6
22470	HEART OF WICKER LARGE	£6,346	1,989	18
84879	ASSORTED COLOUR BIRD ORNAMENT	£5,997	4,071	26

The most consistent contributor across the flagged days is **StockCode 23843** (“PAPER CRAFT, LITTLE BIRDIE”), bought on a single very large wholesale invoice. Two implications:

1. **No removal.** The line is a legitimate transaction by a real customer. Dropping it would silently bias the segment definitions and break reproducibility against the published UCI table.
2. **Profile with and without.** `04_profiles.Rmd` reports the top products per cluster *both* including and excluding 23843 so that one customer's single bulk order does not overstate the assortment of the segment they sit in.